

PROJECT 09 REPORT

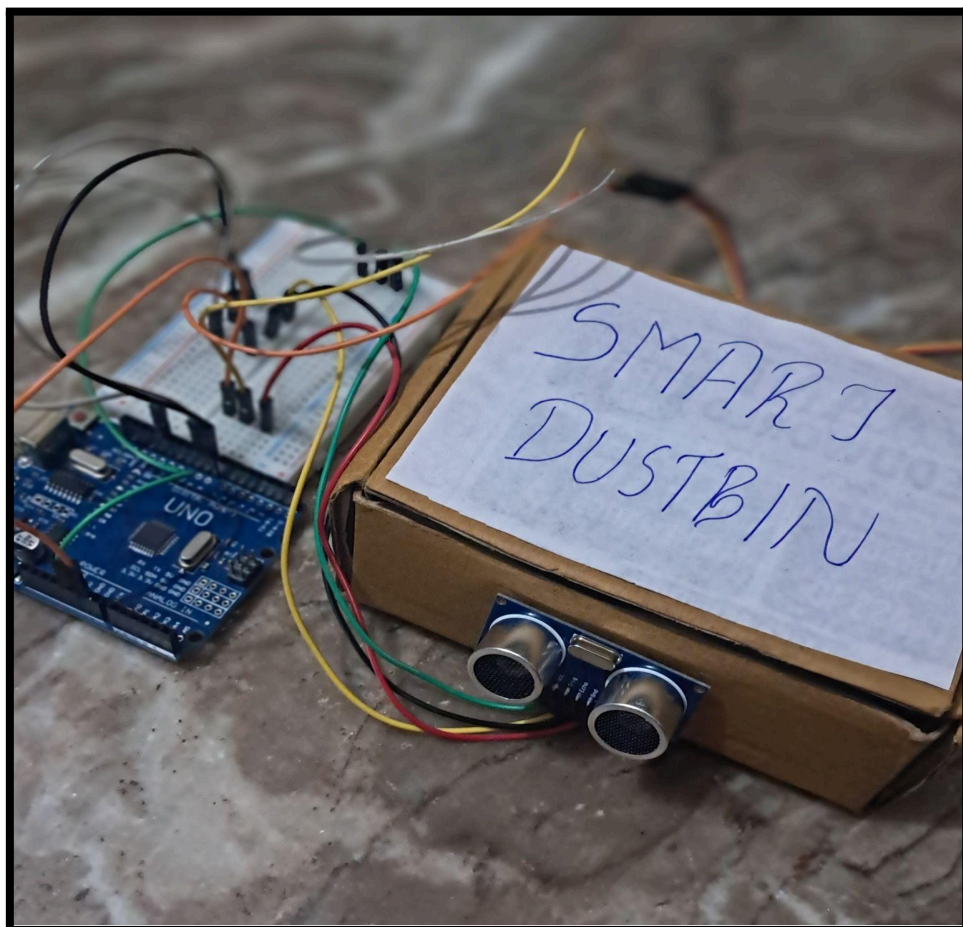
Smart Automatic Dustbin using HC-SR04 Ultrasonic Sensor and SG90 Servo Motor

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Duration : 1 day



1. Objective

The objective of this project was to develop a touchless smart dustbin using HC-SR04 ultrasonic sensor and SG90 servo motor.

The system automatically opens the dustbin lid when an object is detected within a predefined distance range and closes the lid when the object moves outside the range.

2. Components Used

Component	Quantity
Arduino Uno R3	1
HC-SR04 Ultrasonic Sensor	1
SG90 Micro Servo Motor	1
Breadboard	1
Jumper Wires	Multiple
Cardboard Box	1
Ice Cream Stick	1
Double Sided Tape	As need
USB Cable	1

3. Software Used

- Arduino IDE
- Servo Library (Servo.h)
- Windows Laptop

4. Theory

A smart dustbin is an automatic waste disposal system that opens its lid when a user brings their hand or an object near the sensor which eliminates the need for physical contact.

The HC-SR04 ultrasonic sensor continuously measures the distance of objects placed in front of it by transmitting ultrasonic waves and measuring the time taken for the echo to return.

Arduino calculates the distance using:

$$\text{Distance} = (\text{Speed of Sound} \times \text{Time of Flight}) / 2$$

The calculated distance is compared with predefined threshold values.

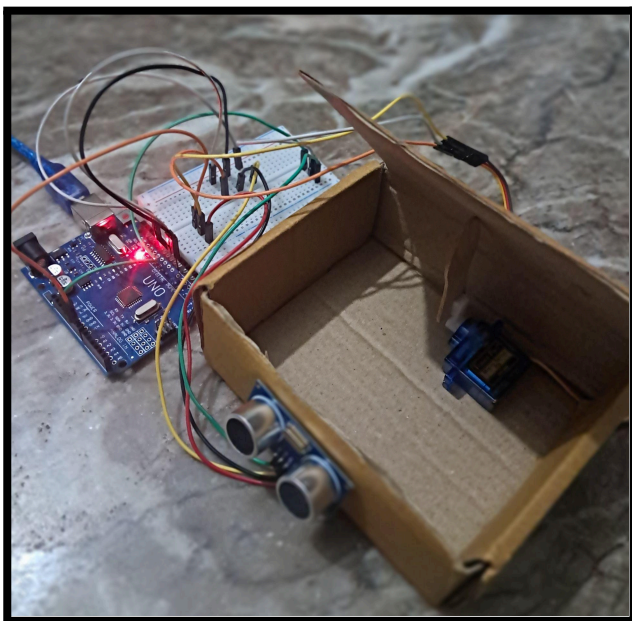
- **Distance < 10 cm** → Arduino commands the servo motor to rotate to **90°**, opening the lid
- **Distance > 12 cm** → Arduino rotates the servo back to **0°**, allowing the lid to close

Using two different threshold values creates a **hysteresis based control system**, preventing the unnecessary opening and closing of the lid when an object remains close to the threshold distance.

The SG90 servo motor acts as the actuator, while the attached ice cream stick functions as a simple mechanical linkage that transfers the servo's rotational motion to the dustbin lid.

5. Tasks Performed

➤ Development of Smart Automatic Dustbin



Video

A complete touchless smart dustbin prototype was developed by integrating the HC-SR04 ultrasonic sensor with an SG90 servo motor.

The ultrasonic sensor was made to continuously monitor the distance of nearby objects.

Object Detected (<10 cm)

- Servo Rotates to 90°
- Ice Cream Stick Pushes Lid
- Dustbin Opens

Object Removed (>12 cm)

- Servo Returns to 0°
- Ice Cream Stick Returns
- Dustbin Closes

Initially, the servo motor was programmed to rotate **180°**, but during testing it was observed that the mechanical linkage over rotated and disturbed the lid mechanism which was later programmed to **90°** resulting in smooth and reliable lid operation.

6. Learning Outcomes

During this project, the following concepts were learned:

- Sensor and Actuator Integration
- Embedded Automation Systems
- HC-SR04 Ultrasonic Distance Measurement
- Servo Motor Position Control
- Threshold Based Decision Making
- Hysteresis Control
- Mechanical Linkage Design
- Prototype Development
- Motion Transfer using Lever Mechanisms

- Practical Embedded System Debugging
- Mechanical and Electronic System Integration
- Rapid Prototype Construction

7. Problems Faced

- I. The servo motor rotated correctly, but the cardboard lid failed to open reliably. **(The servo position was redesigned and an ice cream stick was attached to the servo horn to create a lever mechanism capable of lifting the lid)**
- II. The cardboard components frequently detached during testing. **(The prototype was repeatedly adjusted, repositioned and reinforced until stable operation was achieved)**
- III. The servo initially rotated through 180° , causing the linkage to move beyond the required range. **(The servo angle was reduced from 180° to 90° , producing smooth and controlled lid movement)**

8. Observations

- The HC-SR04 successfully detected nearby objects and triggered automatic lid movement.
- The servo motor reliably opened and closed the lid after mechanical adjustments.
- Using 90° rotation produced better mechanical performance than 180° rotation.

- Hysteresis prevented repeated opening and closing near the threshold distance.
- Hard, flat objects were detected more quickly than soft or irregular objects.
- Mechanical design played an equally important role as programming in the successful operation of the system.
- The completed prototype successfully demonstrated touchless waste disposal.

9. Applications and Future Scope

The concepts learned in this project are used in Touchless Public Waste Bins, Hospital Waste Disposal Systems, Automatic Doors and Contactless Automation Systems.

Possible future upgrade includes OLED Display for System Status, Motion Sensor Integration and Automatic Sanitizer Dispenser Combination Improved Mechanical Enclosure using 3D Printing.

10. Conclusion

This project successfully demonstrated the development of a touchless smart automatic dustbin using an HC-SR04 ultrasonic sensor and an SG90 servo motor.

The project provided practical experience in combining sensors, actuators, programming logic, and mechanical design into a complete embedded system and represents the first **sensor actuator integrated automation system** in the portfolio.